

Joint Reproduction Plan: Noise-Tolerant and Erasure-Biased Rydberg Gates

Noise-Tolerant Quantum Control Optimization

June 22, 2026

- Zhang et al., *Logical Qubits with Erasure Conversion Using Metastable Neutral Atoms* [1].
- Jandura, Thompson, and Pupillo, *Optimizing Rydberg Gates for Logical-Qubit Performance* [2].
- Joint goal: connect robust physical Rydberg gates, erasure conversion, and logical-level performance.

Project Structure

- **Physical-gate layer:** reproduce blockade-limit Rydberg CZ dynamics and robust-pulse sensitivity from Jandura et al.
- **Erasure-conversion layer:** model metastable ^{171}Yb erasure detection and biased error channels from Zhang et al.
- **Logical layer:** compare physical infidelity, erasure bias, and small-code logical performance after the physical channel is validated.
- **Current notebook:** executable baseline for the physical-gate Hamiltonian and error scans.

Each atom is modeled as a three-level system:

$$|0\rangle, \quad |1\rangle, \quad |r\rangle.$$

- Qubit information is stored in $|0\rangle, |1\rangle$.
- A global laser couples only $|1\rangle \leftrightarrow |r\rangle$, with complex Rabi frequency

$$\Omega(t) = \Omega_{\max} e^{i\phi(t)}.$$

- $|0\rangle$ is dark, so states containing $|0\rangle$ only evolve through the other atom if that atom is in $|1\rangle$.
- If both atoms are in $|r\rangle$, the van der Waals blockade shift B makes $|rr\rangle$ off-resonant.

Rotating-Frame Hamiltonian

In the rotating-wave frame, before eliminating $|rr\rangle$,

$$H = \sum_{i=1}^2 \left[\frac{(1 + \epsilon_i)\Omega(t)}{2} |r_i\rangle\langle 1_i| + \text{h.c.} + \Delta_i |r_i\rangle\langle r_i| \right] + B |rr\rangle\langle rr|.$$

- ϵ_i : atom-dependent laser-amplitude or intensity error.
- Δ_i : atom-dependent detuning, e.g. Doppler shift.
- $B \gg \Omega_{\max}$: blockade suppresses double Rydberg excitation.
- The factor $1/2$ is the standard Rabi-frequency convention after the rotating-wave approximation.

Blockade Branches

Because the laser never couples $|0\rangle$, the computational input sectors decouple:

$$H = H_{10} \oplus H_{01} \oplus H_{11}.$$

For example,

$$H_{10} = \frac{(1 + \epsilon_1)\Omega(t)}{2} |r0\rangle\langle 10| + \text{h.c.} + \Delta_1 |r0\rangle\langle r0|,$$

and similarly for H_{01} . For the $|11\rangle$ branch, perfect blockade removes $|rr\rangle$, leaving

$$\{|11\rangle, |r1\rangle, |1r\rangle\}.$$

This is the finite-dimensional model used by the notebook.

Why $W_{\pm}$ Matters

With $\epsilon_i = 0$, the $|11\rangle$ branch can be written in

$$|W_{\pm}\rangle = \frac{|r1\rangle \pm |1r\rangle}{\sqrt{2}}.$$

Then

$$H_{11} = \frac{\sqrt{2}\Omega(t)}{2} |W_+\rangle\langle 11| + \text{h.c.} + \Delta_+ (|W_+\rangle\langle W_+| + |W_-\rangle\langle W_-|) + \Delta_- (|W_+\rangle\langle W_-| + \text{h.c.}),$$

where $\Delta_{\pm} = (\Delta_1 \pm \Delta_2)/2$.

- Common detuning has $\Delta_- = 0$, so it does not mix W_+ and W_- .
- Fig. 2(b) uses $\Delta_1 \neq 0, \Delta_2 = 0$, so $\Delta_- \neq 0$; this coupling must be included.

Imperfections and Error Bias

- Amplitude inhomogeneity: atom-dependent ϵ_i rescales the Rabi coupling.
- Doppler or laser detuning: atom-dependent Δ_i shifts the Rydberg state.
- Optional no-jump decay can be added as $-i\gamma/2$ on Rydberg levels.
- The erasure-conversion line will separate located leakage/erasure-like events from unlocated computational errors.

Fidelity Convention

For a diagonal computational map

$$M = \text{diag}(1, \alpha_{01}, \alpha_{10}, \beta_{11}),$$

the notebook optimizes local phases θ_{01}, θ_{10} and reports process overlap

$$F_{\text{pro}} = \frac{|1 + e^{-i\theta_{01}}\alpha_{01} + e^{-i\theta_{10}}\alpha_{10} - e^{-i(\theta_{01}+\theta_{10})}\beta_{11}|^2}{16}.$$

What Problem AR Pulses Solve

Jandura et al. first isolate amplitude robustness by setting $\Delta_i = 0$ and allowing unknown Rabi-frequency errors [2]:

$$\Omega_i(t) = (1 + \varepsilon_i)\Omega(t).$$

- Physically, ε_i models gate-beam intensity inhomogeneity, slow power drift, or atom-dependent beam-position error.
- A time-optimal pulse can still be very accurate at $\varepsilon_i = 0$, but its return amplitude and acquired phase generally shift linearly with ε_i .
- In an erasure-biased architecture this matters because coherent computational errors are worse than located Rydberg leakage.
- The AR task is therefore not only “make F large”, but make the final computational map insensitive to first order in ε_i .

First-Order Robustness Criterion

The paper expands both the Hamiltonian and each branch state around zero amplitude error:

$$H_q = H_q^{(0)} + \varepsilon H_q^{(1)}, \quad |\psi_q(t)\rangle = |\psi_q^{(0)}(t)\rangle + \varepsilon |\psi_q^{(1)}(t)\rangle + O(\varepsilon^2).$$

- The zero-order condition is still a CZ gate up to one-qubit phases:

$$|\psi_q^{(0)}(\tau)\rangle = e^{i\theta_q}|q\rangle, \quad \theta_{11} - \theta_{10} - \theta_{01} = (2n+1)\pi.$$

- Amplitude robustness adds the stronger endpoint condition

$$|\psi_q^{(1)}(\tau)\rangle = 0 \quad \text{for } q \in \{10, 01, 11\}.$$

- If this holds, the leading coherent endpoint error is quadratic rather than linear in ε_i .

How the Paper Finds AR Pulses

The AR pulse is obtained by minimizing a first-order sensitivity objective:

$$J_{\text{AR}} = 1 - F + \sum_q \langle \psi_q^{(1)}(\tau) | \psi_q^{(1)}(\tau) \rangle.$$

The sensitivity states obey the inhomogeneous equations

$$\dot{\psi}_q^{(0)} = -iH_q^{(0)}\psi_q^{(0)}, \quad \dot{\psi}_q^{(1)} = -iH_q^{(0)}\psi_q^{(1)} - iH_q^{(1)}\psi_q^{(0)}.$$

- The control is represented as a piecewise-constant GRAPE pulse $\Omega(t) = \Omega_j$ on time slot j .
- The optimizer searches for $J_{\text{AR}} = 0$: perfect nominal CZ plus zero first-order amplitude sensitivity.
- The shortest solution found is a phase-only pulse, $\Omega(t) = \Omega_{\text{max}} e^{i\varphi(t)}$, with smooth $\varphi(t)$.

Published AR Pulse Result

The paper reports a critical duration

$$\tau_* \approx 14.32/\Omega_{\max},$$

above which the amplitude-robust objective can be driven to $J_{\text{AR}} = 0$.

- This is longer than the time-optimal CZ pulse, but it keeps the experimentally simple constraint $|\Omega(t)| = \Omega_{\max}$.
- The AR pulse spends an average time $\tau_R = 4.74/\Omega_{\max}$ in the Rydberg state, versus $2.96/\Omega_{\max}$ for the TO pulse.
- In the no-decay diagnostic of Fig. 2(a), the AR pulse keeps $1 - F < 10^{-4}$ for common amplitude errors $|\varepsilon| \leq 0.05$.
- The tradeoff is explicit: more Rydberg exposure, but much flatter coherent error response to laser-amplitude deviations.

Implication for Our Reproduction

The paper's AR construction gives a specific target for this repository:

- The fidelity-only scan is not enough; we must propagate $|\psi_q^{(1)}\rangle$ and minimize its endpoint norm.
- Finite-probe optimization can improve selected endpoints, but it does not prove the derivative condition $|\psi_q^{(1)}(\tau)\rangle = 0$.
- A compact Fourier correction space can diagnose the objective, but the published pulse was found with a much larger GRAPE slot space.
- Therefore the reproduction milestone is paper-level AR first: $\Omega_{\max} T \approx 14.32$, phase-only control, near-unit nominal CZ, and first-order amplitude sensitivity suppressed across all three nontrivial branches.

Initial Numerical Scope

- Reuse the repository's existing time-optimal phase seed from the current ideal global-CZ line.
- Scan common amplitude error and Fig. 2(b)-style single-atom detuning error Δ_1 , with $\Delta_2 = 0$.
- Use this scan as a baseline before adding first-order robustness constraints for AR, DR, ADR, and CADR pulses.
- Store notebook outputs directly in the notebook so the report can cite concrete current-run values.

Current Baseline Output

The saved notebook execution currently gives:

- Dimensionless seed duration: $\Omega_{\max} T = 7.634070$.
- Nominal process fidelity: $F_{\text{pro}} = 0.999993035320$.
- Nominal leakage-like loss: 6.959×10^{-6} .
- Common-amplitude scan minimum fidelity over the starter range: 0.978025007390.
- Fig. 2(b)-style single-atom Δ_1 scan minimum fidelity over the starter range: 0.802681536884.

First Time Optimal/AR/DR Attempt

The notebook now contains a first executable Time Optimal/AR/DR attempt:

- Time Optimal: $F_0 = 0.999993035320$, $1 - F_{\epsilon=+0.04} = 5.221 \times 10^{-3}$,
 $1 - F_{\Delta_1=0.20} = 3.743 \times 10^{-2}$.
- AR first pass: $F_0 = 0.988494437919$, $1 - F_{\epsilon=+0.04} = 4.979 \times 10^{-2}$,
 $1 - F_{\Delta_1=0.20} = 2.263 \times 10^{-2}$.
- DR first pass: $F_0 = 0.9899073209$, sign-reversal $1 - F_{\Delta_1=0.20} = 2.421 \times 10^{-2}$.
- Current diagnosis: the compact Fourier finite-difference proxy is runnable, but it has not yet reproduced the paper's AR/DR robustness.

The notebook now also optimizes over explicit finite noise probes:

- Time Optimal at $\Omega_{\max} T = 14.32$: optimizer seed fidelity 0.999999761162.
- AR, common amplitude $\epsilon = \pm 0.05$: worst endpoint fidelity improves from 0.998262227353 to 0.998552745966.
- AR nominal fidelity after robust balancing: 0.999927393819.
- DR, midpoint sign-reversed $\Delta_1/\Omega_{\max} = \pm 0.20$, $\Delta_2 = 0$: worst endpoint fidelity improves from 0.979043468996 to 0.980303143863.
- DR at the half probe $|\Delta_1|/\Omega_{\max} = 0.10$ remains near 0.99481.

Current Optimization Variables

The notebook optimization is intentionally compact:

$$\phi(t_j) = \phi_{\text{base}}(t_j) + \sum_{m=1}^M a_m \sin\left(\frac{2\pi m t_j}{T}\right) + b_m \cos\left(\frac{2\pi m t_j}{T}\right).$$

- Optimized variables: Fourier coefficients a_m, b_m , and for first-order least-squares also local phases θ_{10}, θ_{01} .
- The laser amplitude is fixed at $|\Omega(t)| = \Omega_{\text{max}}$, matching the paper's phase-only pulse family.
- This is not full GRAPE over all time slots; it searches a much smaller correction space around a seed pulse.

Finite-Probe Objective

The finite-probe pass minimizes a heuristic objective over a small grid of probe errors:

$$J_{\text{probe}} = \sum_{\eta \in \mathcal{P}} (1 - F(\eta))^2 + w_{\text{max}} \max_{\eta \in \mathcal{P}} (1 - F(\eta))^2 + \lambda \|c\|_2^2.$$

- AR probe set: $\eta = \epsilon \in \{0, \pm 0.025, \pm 0.05\}$, with $\epsilon_1 = \epsilon_2$.
- DR probe set: $\eta = \Delta_1, \Delta_2 = 0$, with midpoint sign reversal for the DR candidate.
- This can balance finite endpoints, but it does not enforce vanishing first-order sensitivity.
- Therefore it is a diagnostic optimizer, not the paper's robust-pulse construction.

First-Order Sensitivity Propagation

For a perturbation parameter η , the state expansion is

$$|\psi_q(t)\rangle = |\psi_q^{(0)}(t)\rangle + \eta|\psi_q^{(1)}(t)\rangle + O(\eta^2).$$

The notebook propagates each time slot using Frechet derivatives of the matrix exponential:

$$\left. \frac{\partial}{\partial \eta} \exp[-i(H^{(0)} + \eta H^{(1)})\Delta t] \right|_{\eta=0}.$$

- This avoids finite-difference noise in the first-order states.
- Unit tests compare the propagated $|\psi^{(1)}\rangle$ against central finite differences.
- Atom-resolved detuning now uses separate Δ_1, Δ_2 perturbation traces.

AR versus DR Robust Conditions

AR and DR do not impose the same first-order condition.

- AR follows the paper's amplitude-robust objective:

$$J_{\text{AR}} = 1 - F + \sum_q \langle \psi_q^{(1)} | \psi_q^{(1)} \rangle.$$

The full first-order state is penalized.

- DR follows the paper's projected condition:

$$J_{\text{DR}} = 1 - F + \sum_{j,q} \langle \psi_q^{(1,j)} | (I - |q\rangle\langle q|) | \psi_q^{(1,j)} \rangle.$$

- The component along $|q\rangle$ is allowed for DR because the second half flips the Doppler sign and cancels that first-order phase contribution.

First-Order Sensitivity Objective

The notebook now implements the paper-style $|\psi_q^{(1)}\rangle$ sensitivity-state objective:

- Reusable helper: exact slot-wise first-order propagation via matrix exponential Frechet derivatives, including atom-resolved detuning.
- AR at $\epsilon = \pm 0.05$: worst endpoint fidelity improves from 0.995662275603 to 0.996558440020.
- AR sensitivity norm drops from 9.636384×10^1 to 5.740975, with $F_0 = 0.999991477077$.
- Corrected DR diagnostic at $\Delta_1/\Omega_{\max} = \pm 0.20$, $\Delta_2 = 0$: current compact full-pulse search drops from Time Optimal worst endpoint 0.976711957540 to 0.906992297543.
- Diagnosis: the remaining Fig. 2(b) gap is pulse construction; next implement the repeated CRz($\pi/2$) half-pulse GRAPE used in the paper.

Why the Current Optimization Underperforms

The corrected model exposes a real algorithmic gap:

- **Pulse structure mismatch:** the paper optimizes a $\text{CRz}(\pi/2)$ half-pulse and applies it twice with opposite Doppler sign; the notebook still searches one compact full-pulse correction.
- **Restricted basis:** 10 Fourier coefficients cannot span the same space as slot-wise or documented-basis GRAPE.
- **No analytic GRAPE gradient:** least-squares over compact coefficients is slower and more prone to local minima.
- **Seed mismatch:** our Time Optimal seed is from the repository's current line, not the paper's released optimized pulse samples.
- **Weight sensitivity:** large zero-order weights protect nominal fidelity but can suppress movement toward robust manifolds.

Next Reproduction Milestones

- ➊ Replace the compact first-order search with full slot-wise or documented-basis GRAPE gradients.
- ➋ Reproduce the published AR/DR pulse families at paper-level fidelity.
- ➌ Extend the Doppler sign-reversal implementation to ADR/CADR pulse families.
- ➍ Decompose physical errors into erasure-like and computational components.
- ➎ Add Zhang-style erasure-conversion checks for metastable ^{171}Yb .
- ➏ Only after the physical channel is validated, add a small logical-error-rate surrogate.

- [1] Bichen Zhang et al. “Logical Qubits with Erasure Conversion Using Metastable Neutral Atoms”. In: *Nature Physics* 22.6 (June 2026), pp. 910–916. ISSN: 1745-2481. DOI: 10.1038/s41567-026-03309-0. URL: <https://www.nature.com/articles/s41567-026-03309-0> (visited on 06/22/2026).
- [2] Sven Jandura, Jeff D. Thompson, and Guido Pupillo. “Optimizing Rydberg Gates for Logical-Qubit Performance”. In: *PRX Quantum* 4.2 (June 5, 2023), p. 020336. ISSN: 2691-3399. DOI: 10.1103/PRXQuantum.4.020336. URL: <https://link.aps.org/doi/10.1103/PRXQuantum.4.020336> (visited on 06/22/2026).